

Final Exam Solutions:

1. (Counting and Discrete Probability; 15 pts.)

Suppose 26 uppercase letters, 26 lowercase letters, and 10 digits (so 62 characters in all) can be used to form passwords. For each part below, find how many possible passwords of length six can be formed under the conditions specified. Please do not compute a final number – i.e., leave your answers in terms of exponents, factorials, combinations, etc.

- (a) The first four characters must be letters, the last two must be digits, and repetition of characters *is* allowed.
 - (b) The first four characters must be letters, the last two must be digits, but repetition of characters is *not* allowed.
 - (c) There must be exactly two digits, but they can be in any of the six positions. Repetition of characters *is* allowed.
 - (d) There must be exactly two digits, but they can be in any of the six positions. Repetition of characters is *not* allowed.
 - (e) Suppose a password of length six is chosen at random – i.e., the character in each position is drawn uniformly from the 62 possible characters and independent of all other characters. Let A be the event that a randomly chosen password has the first four characters being non-repeating letters, and the last two characters being non-repeating digits. Let B be the event that a randomly chosen password has four letters and two digits, but with no other restrictions. Find $\mathbb{P}(A|B)$. You can use your results of the previous parts.
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(a) Sub-experiments 1, 2, 3, 4: (Each) choosing 1 char. from 52 total letters
 $\leadsto 52^4$ possibilities

Sub-experiments 5, 6: (Each) choosing 1 char. from 10 digits
 $\leadsto 10^2$ possibilities

Total # of pwd's: $52^4 \cdot 10^2$

(b) Sub-exp 1: Choose 4 out of 52 letters (order matters)
 $\leadsto P(52, 4)$ possibilities

Sub-exp. 2: Choose 2 out of 10 digits (order matters)
 $\leadsto P(10, 2)$ possibilities

Total # of pwd's: $P(52, 4) \cdot P(10, 2)$

(c) Sub-exp. 1: Choose 2 out of 6 positions to have digits

$$\leadsto \binom{6}{2} = C(6, 2) \text{ possibilities}$$

Sub-exp. 2: Choose 4 letters & 2 digits, repetition allowed

$$\leadsto 54^4 \cdot 10^2, \text{ from part (a)}$$

$$\text{Total \# of pwd's: } \binom{6}{2} 54^4 \cdot 10^2$$

(d) Sub-exp. 1: Choose 2 out of 6 positions to have digits

$$\leadsto \binom{6}{2} = C(6, 2) \text{ possibilities}$$

Sub-exp. 2: Choose 4 letters & 2 digits, repetition not allowed

$$\leadsto P(54, 4) \cdot P(10, 2), \text{ from part (b)}$$

$$\text{Total \# of pwd's: } \binom{6}{2} P(54, 4) \cdot P(10, 2)$$

(e) Note that $A \subset B$, so $A \cap B = A$. Also, the total # of possible pwd's is 62^6

By Bayes' Thm:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$= \frac{P(A)}{P(B)} \quad \left. \begin{array}{l} \text{By } A \subset B \\ \text{using (b)} \end{array} \right\}$$

$$P(A) = \frac{P(54, 4) \cdot P(10, 2)}{62^6}$$

$$P(B) = \frac{\binom{6}{2} 54^4 10^2}{62^6}$$

$$P(A|B) = \frac{P(54, 4) \cdot P(10, 2)}{\binom{6}{2} 54^4 10^2}$$

2. (Mean, Variance, and CDF; 15 pts.)

Let X be a discrete random variable with $P(X = 1) = \alpha$ and $P(X = -1) = 1 - \alpha$. Let Y be a random variable with a standard normal distribution, and let $\Phi(\cdot)$ denote its CDF. Assume that X and Y are independent. Let $Z = X|Y|$. That is, Z is equal to X times the absolute value of Y .

- Find the expected value $\mathbb{E}[Z]$ and variance $\text{Var}[Z]$ of the random variable Z . You may use the fact that $\mathbb{E}[|Y|] = \sqrt{2/\pi}$.
- Find $F_Z(z)$ (i.e., the CDF of Z) in terms of $\Phi(\cdot)$. (Hint: consider separately the cases $z \leq 0$ and $z > 0$ and get an expression for each case.)
- Find a simple expression for $F_Z(z)$ in terms of $\Phi(\cdot)$ for the special case when $\alpha = 1/2$.

$$\begin{aligned}
 \text{(a)} \quad \mathbb{E}[Z] &= \mathbb{E}[X \cdot |Y|] \\
 &= \mathbb{E}[X] \cdot \mathbb{E}[|Y|] \quad \text{by independence} \\
 &= [\alpha \cdot 1 + (1-\alpha) \cdot (-1)] \cdot \sqrt{\frac{2}{\pi}} \\
 \boxed{\mathbb{E}[Z] &= (2\alpha - 1) \sqrt{\frac{2}{\pi}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{Var}(Z) &= \mathbb{E}[Z^2] - (\mathbb{E}[Z])^2 \\
 &= \mathbb{E}[X^2 \cdot Y^2] - (2\alpha - 1)^2 \frac{2}{\pi} \quad \text{plugging in from above} \\
 &= \underbrace{\mathbb{E}[X^2]}_{=1} \cdot \underbrace{\mathbb{E}[Y^2]}_{\text{by independence}} - (2\alpha - 1)^2 \frac{2}{\pi} \\
 &\quad \text{Var}(Y) + (\mathbb{E}[Y])^2 = 1 + 0 = 1 \\
 &\quad Y \sim N(0,1)
 \end{aligned}$$

$$\boxed{\text{Var}(Z) = 1 - (2\alpha - 1)^2 \frac{2}{\pi}}$$

(b) If $z < 0$, then it must be that $X = -1$

So:

$$\begin{aligned}
 P(Z \leq z) &= P(X = -1, \text{ and } -|Y| \leq z) \\
 &= P(X = -1) \cdot P(|Y| \geq -z) \quad \text{by independence} \\
 &= (1 - \alpha) \cdot 2 \cdot P(Y \leq z) \quad \text{by symmetry of } N(0,1)
 \end{aligned}$$

$$\underline{F_Z(z) = 2(1 - \alpha) \Phi(z)}, \text{ for } z < 0$$

If $z \geq 0$, then either:

- $X = -1$, and we have any Y
- $X = +1$, and $|Y| \leq z$

$$P(Z \leq z) = P(X = -1) + P(X = +1) \cdot \underbrace{P(|Y| \leq z)}_{= 1 - 2 \cdot \Phi(-z)}$$

$$= (1 - \alpha) + \alpha (1 - 2\Phi(-z))$$

$$F_Z(z) = \underline{1 - 2\alpha \Phi(-z)}, \text{ for } z \geq 0$$

$$\therefore F_Z(z) = \begin{cases} 1 - 2\alpha \Phi(-z) & \text{if } z \geq 0 \\ 2(1 - \alpha)\Phi(z) & \text{if } z < 0 \end{cases}$$

(c) Plug in $\alpha = 1/2$ to part b:

$$F_Z(z) = \begin{cases} 1 - \Phi(-z) & \text{if } z \geq 0 \\ \Phi(z) & \text{if } z < 0 \end{cases}$$

Since $1 - \Phi(-z) = \Phi(z)$, we obtain

$$\boxed{F_Z(z) = \Phi(z)}!$$

3. (Point Estimation; 15 pts.)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Geom}(\theta)$ be independent and identically distributed random variables from a Geometric distribution with success probability $\theta \in [0, 1]$. Namely, each X_i counts the number of independent trials, each with success probability θ , up to and including the first success (not the number of failures before the first success).

For example, for $n = 2$ we might obtain the following realizations:

- $x_1 = 3$ (First two trials were a failure, third was a success)
- $x_2 = 5$ (First four trials were a failure, fifth was a success)

In general, if $x_i = k$, then we obtained $k - 1$ failures followed by a success on the k -th trial.

The pmf of each of $X_1, \dots, X_n$ is given by:

$$p(x; \theta) = \begin{cases} (1 - \theta)^{x-1} \theta & \text{for all } x = 1, 2, 3, \dots \\ 0 & \text{otherwise.} \end{cases}$$

In this problem, we will build and analyze the maximum likelihood estimator for θ .

- Derive the likelihood function $\mathcal{L}(\theta; X_1, \dots, X_n)$, and the log-likelihood function.
- Compute the maximum likelihood estimator $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ for θ .
- Is $\hat{\theta}_{MLE}(X_1, \dots, X_n)$ a biased estimator of θ ? Be sure to justify your answer (a proof is not necessary, but is certainly acceptable).

You may use (without proof) that $\mathbb{E}[X_1] = \frac{1}{\theta}$.

$$(a) \mathcal{L}(\theta; X_1, \dots, X_n) = \prod_{i=1}^n f(x_i; \theta) \quad \text{by def'n}$$

$$= \prod_{i=1}^n (1 - \theta)^{x_i - 1} \cdot \theta$$

$$\mathcal{L}(\theta; X_1, \dots, X_n) = \theta^n (1 - \theta)^{(\sum_{i=1}^n x_i) - n}$$

$$\log \mathcal{L}(\theta; X_1, \dots, X_n) = n \log(\theta) + \left[\left(\sum_{i=1}^n x_i \right) - n \right] \log(1 - \theta)$$

$$(b) \text{ Solve for } \frac{\partial}{\partial \theta} \log(\mathcal{L}(\hat{\theta}_{MLE})) = 0:$$

$$0 = \frac{n}{\hat{\theta}_{MLE}} + \left[\left(\sum_{i=1}^n x_i \right) - n \right] \cdot \frac{-1}{1 - \hat{\theta}_{MLE}}$$

$$\Rightarrow \hat{\theta}_{MLE} = \frac{1}{\frac{1}{n} \sum_{i=1}^n x_i}$$

(c) Yes $\hat{\theta}_{MLE}$ is biased.

$$E[\hat{\theta}_{MLE}] = E\left[\frac{1}{\bar{X}_n}\right]$$

This will not equal $E\left[\frac{1}{\bar{X}_n}\right] = \frac{1}{\theta} = \theta$

4. (Confidence Intervals; 15 pts.)

Let $X_1, \dots, X_n$ be iid random variables whose probability density function is given by

$$f(x; \theta) = \begin{cases} \frac{\theta}{x^{\theta+1}}, & \forall x \geq 1 \\ 0 & \text{otherwise,} \end{cases}$$

where the parameter $\theta > 1$ is unknown. Assume that $n > 40$.

(a) Compute $E[X_1]$ and $\text{Var}(X_1)$. Recall that we assumed $\theta > 1$.

(b) Let $\bar{X}_n$ denote the mean of $X_1, \dots, X_n$, and $s^2(X_1, \dots, X_n) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$ an estimator for the population variance. For $\alpha \in (0, 1)$, let $z_{\alpha/2}$ be the $\alpha/2$ critical value of a standard normal random variable as usual.

For what range of θ does the following statement hold?

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\sqrt{n} \left(\bar{X}_n - \frac{\theta}{\theta-1}\right)}{s(X_1, \dots, X_n)} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Hint: Think carefully about the necessary conditions for applying the Central Limit Theorem.

(c) Use the expression in part (b) to construct a two-sided $1 - \alpha$ confidence interval for θ . Be careful to notice that θ is **not** the mean of $X_1, \dots, X_n$.

$$\begin{aligned} (a) \quad E[X] &= \int_1^{\infty} x \cdot \frac{\theta}{x^{\theta+1}} dx \\ &= \theta \cdot \int_1^{\infty} x^{-\theta} dx = \theta \cdot \left[\frac{1}{1-\theta} x^{1-\theta} \right]_1^{\infty} \\ &= \theta \cdot \left[0 - \frac{1}{1-\theta} \cdot 1 \right] \\ &= \frac{\theta}{\theta-1} \end{aligned}$$

$$\text{Var}(X) = \underbrace{E[X^2]} - (E[X])^2$$

$$\hookrightarrow = \int_1^{\infty} x^2 \frac{\theta}{x^{\theta+1}} dx = \theta \int_1^{\infty} x^{1-\theta} dx$$

$$= \theta \cdot \left[\frac{1}{2-\theta} x^{2-\theta} \right]_1^{\infty} = \frac{\theta}{\theta-2} \quad \parallel \text{ only converges when } \theta > 2$$

else $E[X^2] = \infty$

$$= \frac{\theta}{\theta-2} - \left(\frac{\theta}{\theta-1} \right)^2 \quad (\text{for } \theta > 2)$$

$$\text{Var}(X) = \frac{\theta}{(\theta-2)(\theta-1)^2} \quad (\text{for } \theta > 2)$$

$$\text{So: } \boxed{\text{Var}(X) = \begin{cases} \infty & \text{if } \theta \leq 2 \\ \frac{\theta}{(\theta-2)(\theta-1)^2} & \text{otherwise} \end{cases}}$$

(b) For the CLT to apply, we need:

i) iid random variables ✓

ii) $\text{Var}(X_i) < \infty$ ← Only occurs when $\theta > 2$, as above

∴ The statement only holds for $\theta > 2$

(c) Re-arranging the given expression:

$$P\left(-z_{\frac{\alpha}{2}} \leq \frac{\sqrt{n}(\bar{X}_n - \frac{1}{1-1/\theta})}{s} \leq z_{\frac{\alpha}{2}}\right) = 1 - \alpha$$

↓

$$P\left(\bar{X}_n - z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}} \leq \frac{1}{1-1/\theta} \leq \bar{X}_n + z_{\frac{\alpha}{2}} \frac{s}{\sqrt{n}}\right) = 1 - \alpha$$

After more algebra:

$$P\left(\frac{1}{1 - \frac{1}{\bar{X}_n + z_{\frac{\alpha}{2}} s/\sqrt{n}}} \leq \theta \leq \frac{1}{1 - \frac{1}{\bar{X}_n - z_{\frac{\alpha}{2}} s/\sqrt{n}}}\right) = 1 - \alpha$$

5. (Commuting Distances; 15 pts.)

The U.S. Census Bureau determines how far the head of a household has to commute to work using a nation-wide survey. Based on an *extremely* large sample, the national average commute distance is 13 miles.

A real estate office wants to make a similar survey in a certain town. A random sample of 400 households is chosen, the occupants are interviewed, and it is determined that on the average, the heads of the sampled households commuted on average 13.7 miles to work, with a standard deviation of 12.0 miles.

- Construct a lower one-sided 95% confidence interval for the mean commute time for the head of households in this town. Justify any approximations or assumptions you may have used in your computation.
- On average, do the heads of households in the town have to commute farther than their national counterparts? Be sure to: i) state the relevant null and alternative hypotheses, ii) perform an appropriate test at significance level 5%, and iii) ensure that you report *both* the outcome of the test *and* the associated p-value.

You may assume that the national average commute distance is 13 miles.

(a) Pop. dist'n is not necessarily normal, but the sample size $n=400$ is large
 $\leadsto X_1, \dots, X_{400}$ iid, with unknown mean μ , var. σ^2

Assume $\sigma^2 < \infty$ (so CLT can be applied).

Hence, $\frac{\sqrt{n}(\bar{X}_n - \mu)}{s(X_1, \dots, X_n)} \overset{\text{approx.}}{\sim} N(0, 1)$

$$\Rightarrow P\left(\frac{\sqrt{n}(\bar{X}_n - \mu)}{s(X_1, \dots, X_n)} \leq z_\alpha\right) = 1 - \alpha$$

Plug in: $n=400$, $\bar{X}_n=13.7$, $s=12.0$, $\alpha=5\%$

$$\boxed{95\% \text{ CI: } [13.7 - 1.65 \cdot \frac{12}{20}, \infty)}$$

(b) $H_0: \mu = 13 = \text{national avg.}$

$H_A: \mu > 13$

$$T(X_1, \dots, X_n) = \frac{\sqrt{n}(\bar{X}_n - 13)}{s(X_1, \dots, X_n)} \leftarrow N(0, 1) \text{ under } H_0$$

$$R = [z_\alpha, \infty)$$

Test Outcome: Plug in $n=400$, $\bar{X}_n=13.7$, $s=12.0$, $\alpha=5\%$

$$T(x_1, \dots, x_n) = 1.17 \notin R$$

$\leadsto D_0$ not reject H_0

$$\boxed{p\text{-Val.} = P(T(\dots) \geq 1.17) = (1 - \Phi(1.17)) \approx 12.1\%}$$

6. (Two-Sample Tests with Dependent Samples; 15 pts.)

A clinical trial measures the blood sugar of $n > 40$ identical diabetic patients before and after a new drug is taken. Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu_X, \sigma_X^2)$ denote the blood sugar of patients pre-treatment, and $Y_1, \dots, Y_n \stackrel{iid}{\sim} N(\mu_Y, \sigma_Y^2)$ denote the post-treatment measurements.

Note that X_i and Y_i correspond to the *same* patient, before and after treatment.

Assume the following:

- The population variances σ_X^2 and σ_Y^2 are known.
- Both $\mu_X \neq 0$ and $\mu_Y \neq 0$.
- X_i and Y_i are **not independent**, and are related by $Y_i = \beta X_i + \epsilon_i$, for some constant $\beta \in \mathbb{R}$, and $\epsilon_i \sim N(0, 1)$ independent of X_i .

We are interested in testing the following hypotheses $H_0 : \mu_X = \mu_Y$, $H_A : \mu_X \neq \mu_Y$. Note that the null hypothesis $H_0 : \mu_X = \mu_Y$ corresponds to $\beta = 1$.

- Show that $\text{Var}(X_i - Y_i) = (1 - \beta)^2 \sigma_X^2 + 1$.
- Express σ_Y^2 in terms of σ_X^2 , and determine which value(s) of β yield $\text{Var}(X_i - Y_i) = \sigma_X^2 + \sigma_Y^2$?
- Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, and $\bar{Y}_n = \frac{1}{n} \sum_{i=1}^n Y_i$ denote the sample means as usual. Consider the following test statistic:

$$T_1(X_1, \dots, X_n, Y_1, \dots, Y_n) = \frac{\sqrt{n}(\bar{X}_n - \bar{Y}_n)}{\sqrt{\sigma_X^2 + \sigma_Y^2}},$$

A friend claims that $T_1(\dots)$, paired with the rejection region $R = (-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, \infty)$, yields a hypothesis test of size α . Why is this claim incorrect?

Hint: Think of the relationship in part (a), and the assumptions needed for a two-sample hypothesis test.

- Instead, you propose a hypothesis test based on the difference in each patient's blood sugar levels around the treatment. That is, let $Z_i = X_i - Y_i$, and define

$$T_2(X_1, \dots, X_n, Y_1, \dots, Y_n) = \sqrt{n} \bar{Z}_n,$$

where $\bar{Z}_n = \frac{1}{n} \sum_{i=1}^n Z_i$ is defined as usual.

What is the distribution of $T_2(\dots)$ under the null hypothesis $H_0 : \mu_X = \mu_Y$? Determine if this test statistic gives a proper test of size α with the same rejection region R from part (c).

$$\begin{aligned} (a) \quad \text{Var}(X_i - Y_i) &= \text{Var}(X_i - (\beta X_i + \epsilon_i)) \\ &= \text{Var}((1-\beta)X_i - \epsilon_i) \\ &= \text{Var}((1-\beta)X_i) + \text{Var}(\epsilon_i) \end{aligned}$$

By independence

$$\boxed{\text{Var}(X_i - Y_i) = (1-\beta)^2 \sigma_X^2 + 1}$$

$$\begin{aligned} (b) \quad \sigma_Y^2 &= \text{Var}(Y_i) = \text{Var}(\beta X_i + \epsilon_i) \\ &= \text{Var}(\beta X_i) + \text{Var}(\epsilon_i) \end{aligned}$$

by independence

$$\boxed{\sigma_Y^2 = \beta^2 \sigma_X^2 + 1}$$

$$\text{For: } \text{Var}(X_i - Y_i) = \sigma_X^2 + \sigma_Y^2$$

$$\begin{aligned} (1-\beta)^2 \sigma_X^2 + 1 &= \sigma_X^2 + \beta^2 \sigma_X^2 + 1 \\ &= (1+\beta^2) \sigma_X^2 + 1 \end{aligned}$$

$$\Rightarrow (1-\beta)^2 = 1 + \beta^2$$

$$\Rightarrow \underline{\underline{\beta = 0}}$$

$$\boxed{\therefore \text{Only for } \beta = 0 \text{ does } \text{Var}(X_i - Y_i) = \sigma_X^2 + \sigma_Y^2}$$

(c) Under H_0 , we must have $\beta = 1$.

But when $\beta = 1$, $\text{Var}(X_i - Y_i) \neq \sigma_X^2 + \sigma_Y^2$

Hence, $T(X_1, \dots, X_n, Y_1, \dots, Y_n) \not\sim N(0, 1)$, under H_0 , and the resulting test will not have size α .

Intuitively, this occurs because the two samples

$X_1, \dots, X_n$ and $Y_1, \dots, Y_n$ are not independent from each other, unless $\beta = 0$.

$$\begin{aligned} \text{(d) Under } H_0, \beta &= 1. \text{ So } Z_i = X_i - Y_i \\ &= X_i - \beta \cdot X_i - \varepsilon_i \\ &= -\varepsilon_i \end{aligned}$$

$$\Rightarrow Z_i \sim N(0, 1)!$$

So, under H_0 , $Z_1, \dots, Z_n \stackrel{iid}{\sim} N(0, 1)$

By CLT, under H_0 , $T_2(X_1, \dots, X_n, Y_1, \dots, Y_n) = \frac{\sqrt{n}(\bar{Z}_n - 0)}{1} \sim N(0, 1)$ also.

$$\boxed{\therefore \text{With } R = (-\infty, -z_{\frac{\alpha}{2}}] \cup [z_{\frac{\alpha}{2}}, \infty), T_2(\dots) \text{ gives a test of size } \alpha!}$$